CAPSTONE 31 — Final Demo & Evaluator Walkthrough

10-Minute Presentation Script, Rubric Mapping & Viva Q&A

1. Final Demo Walkthrough & 100-Mark Evaluation Guide

Use this structured 10-minute presentation and demonstration script during the final capstone evaluation to walk faculty assessors through all 12 project requirements inside the single project folder.

Time Window	Final Demo Stage	Folder File Demonstrated	Key Evidence Shown to Evaluator
Min 0:00–2:00	**Problem, Audience & Research** (Tasks 1–4)	`01_Research_Report.pdf` `02_Git_and_CI_Checklist.pdf`	Explain why 2nd/3rd-year CSE teams need GitHub Flow over Gitflow; show the 6-AI-tool research matrix and the 3-phase team checklist grounded in Pro Git.
Min 2:00–4:00	**Visual Architecture & Slides** (Tasks 5 & 8a)	`03_Git_Guide_and_Workflow_Diagram.pdf` `07_Presentation.pptx`	Walk through the 6-slide presentation deck and the end-to-end Team Workflow Diagram showing local hooks, PR templates, and CI gates.

Min 4:00–7:00	**Live Terminal & CI Execution** (Tasks 6, 7 & 9)	`04_Sample_Repo/`
		`05_CI_Workflow.yml`
		`08_Demo.pdf`
			Run live terminal commands showing `scripts/commit-msg` blocking a bad commit header, passing a valid Conventional Commit, and passing `lint_check.py` + 5 unit tests.

Min 7:00–9:00	**AI Verification, Prompts & Ethics** (Tasks 10–12)	`09_Prompt_Library.pdf`
		`10_Verification_Report.pdf`
		`11_Before_After_Analysis.pdf`
			Demonstrate the 10 R-C-T-V prompts, the 5 caught AI hallucinations (`checkout@v2`, `::set-output`), the 3 Ethics Pillars, and the 86% defect reduction chart.

Min 9:00–10:00	**Team Assessment & Viva Q&A** (Task 8b)	`06_Quiz_and_Answer_Key.pdf` `12_Final_Demo.pdf`	Review the 10-question diagnostic quiz and answer key, and address evaluator viva questions on client-side hooks vs. server-side CI.

2. Evaluator Viva Questions & Model Technical Answers

- **Q1: Why can't a team rely solely on the local `commit-msg` hook without GitHub Actions CI?**
  Model Answer: Local Git hooks reside in `.git/hooks/` on each student's laptop and are not automatically cloned by Git, or can be bypassed using `git commit --no-verify`. Server-side GitHub Actions CI runs in a clean isolated Ubuntu VM on every Pull Request and is enforced by GitHub Branch Protection rules (GitHub Docs).
- **Q2: How does AI help student teams without replacing their engineering judgment?**
  Model Answer: AI accelerates onboarding by translating abstract Git DAG pointer mechanics into interactive analogies and generating initial YAML/test scaffolding, while human engineering judgment and official documentation verification catch outdated syntax, security flaws, and edge cases.